GATE ME 90 Days Challenge

Day 38/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Thermodynamics	Second Law & Entropy	67 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on the Kelvin–Planck and Clausius statements, heat engines, refrigerators and heat pumps, the Carnot cycle and reversed Carnot cycle, the Clausius inequality, entropy change of solids, liquids and ideal gases, T–s diagrams, isentropic efficiency and entropy generation. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE. **Today's test:** open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Which statement about the Kelvin–Planck (K–P) and Clausius statements of the second law is correct?

(A) The two statements are independent; one may be violated without the other

(B) A device violating the Clausius statement can be combined with an ordinary engine to violate the K–P statement

(C) The Clausius statement applies only to refrigerators, so it cannot be compared with the K–P statement

(D) The K–P statement follows from the first law

Q2 NAT

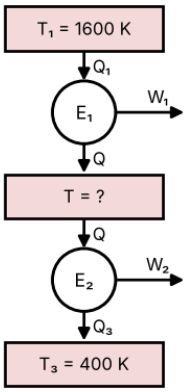
A reversible heat engine works between reservoirs at 900 K and 300 K and receives 1800 kJ of heat per cycle. The work output per cycle (in kJ) is _____

Q3 NAT

A refrigerator keeps its freezer compartment at −13 °C while the kitchen is at 37 °C. If heat leaks into the freezer at 2 kW, the minimum power needed to run the refrigerator (in W, round off to one decimal place) is _____

Q4 MCQ

Two reversible engines E_1 and E_2 work in series between 1600 K and 400 K as shown, E_2 receiving all the heat rejected by E_1 at the intermediate temperature T. If the two engines have equal thermal efficiencies, T is



(A) 1000 K
(B) 1200 K
(C) 800 K
(D) 600 K

Q5 NAT

A 2 kg copper block (c = 0.4 kJ/kg·K) at 500 K is dropped into a large lake at 300 K and comes to equilibrium. The total entropy generated (in kJ/K, round off to three decimal places) is _____

Q6 MCQ

An engine receives 500 kJ from a reservoir at 1000 K and rejects 300 kJ to a reservoir at 500 K in each cycle. Applying the Clausius inequality, the engine is

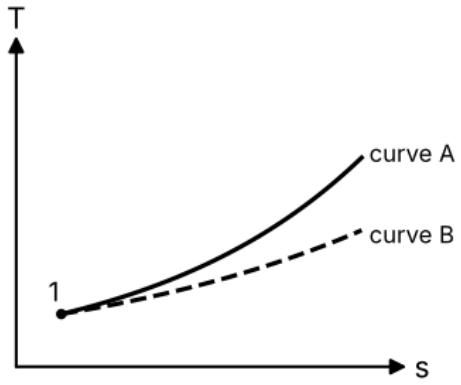
(A) irreversible but possible
(B) reversible
(C) impossible
(D) a heat pump

Q7 NAT

Air (c_v = 0.718 kJ/kg·K) in a rigid tank is heated from 300 K to 900 K. The entropy change per kg (in kJ/kg·K, round off to three decimal places) is _____

Q8 MCQ

Curves A and B on the T–s diagram both start at state 1 of an ideal gas; one is a constant-pressure line and the other a constant-volume line. Which statement is correct?



(A) A is the constant-pressure line, because $c_p > c_v$

(B) Both lines have the same slope at state 1

(C) A is the constant-pressure line, because its slope is T/c_p

(D) A is the constant-volume line, because its slope T/c_v is greater than T/c_p

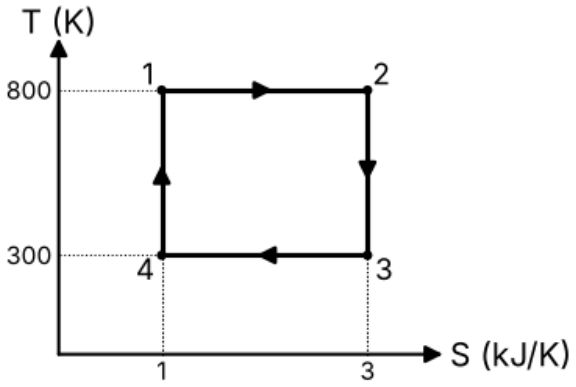
Q9 NAT

Air ($\gamma = 1.4$) expands adiabatically in a turbine from 800 kPa, 1000 K to 100 kPa. The actual exit temperature is 620 K. The isentropic efficiency of the turbine (round off to three decimal places) is

_____

Q10 MCQ

A Carnot cycle 1–2–3–4 is drawn on the T–S diagram shown. The net work per cycle is



(A) 1600 kJ

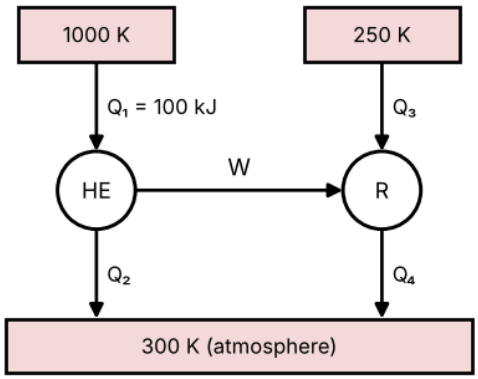
(B) 1000 kJ

(C) 600 kJ

(D) 2200 kJ

Q11 NAT

A reversible engine HE takes $Q_1 = 100$ kJ from a source at 1000 K and rejects heat to the atmosphere at 300 K. All its work drives a reversible refrigerator R that removes Q_3 from a cold space at 250 K and rejects heat to the atmosphere, as shown. Q_3 (in kJ) is _____



Q12 MCQ

Which one of the following processes can be reversible (in the ideal limit)?

(A) Throttling of a gas through a porous plug

(B) Mixing of two different gases

(C) Slow, frictionless adiabatic expansion of a gas behind a piston

(D) Stirring of a liquid in an insulated tank by a paddle wheel

Q13 NAT

Air (an ideal gas, R = 0.287 kJ/kg·K) is throttled steadily from 1 MPa to 200 kPa in an insulated valve; kinetic energy changes are negligible. The entropy generated per kg of air (in kJ/kg·K, round off to three decimal places) is _____

Q14 MCQ

Heat flows steadily at 2 kW through a wall whose inner surface is at 300 K and outer surface at 275 K. The rate of entropy generation within the wall is

(A) 0.606 W/K

(B) 6.67 W/K

(C) 7.27 W/K

(D) zero, since the wall is in steady state

Q15 MCQ

A reversible heat pump working between two reservoirs has a COP of 5. When the same machine is run as a refrigerator between the same two reservoirs, its COP is

(A) 6

(B) 5

(C) 0.2

(D) 4

Answer Key

Q	Type	Answer
Q1	MCQ	(B) A device violating the Clausius statement can be combined with an ordinary engine to violate the K–P statement
Q2	NAT	1200
Q3	NAT	384.6 (accept 382.6 to 386.6)
Q4	MCQ	(C) 800 K
Q5	NAT	0.125 (accept 0.124 to 0.126)
Q6	MCQ	(A) irreversible but possible
Q7	NAT	0.789 (accept 0.785 to 0.793)
Q8	MCQ	(D) A is the constant-volume line, because its slope T/c_v is greater than T/c_p
Q9	NAT	0.848 (accept 0.844 to 0.852)
Q10	MCQ	(B) 1000 kJ
Q11	NAT	350
Q12	MCQ	(C) Slow, frictionless adiabatic expansion of a gas behind a piston
Q13	NAT	0.462 (accept 0.459 to 0.465)
Q14	MCQ	(A) 0.606 W/K
Q15	MCQ	(D) 4

Detailed Solutions

Q1 Answer: (B) A device violating the Clausius statement can be combined with an ordinary engine to violate the K–P statement

Concept: The two statements are equivalent: violating either one implies violating the other.

Let a 'Clausius-violating' device move Q_2 from the cold to the hot reservoir with no work. Pair it with an ordinary engine that takes Q_1 from the hot reservoir and rejects Q_2 to the cold one.

The cold reservoir then has no net exchange, and the combination takes $(Q_1 - Q_2)$ from the hot reservoir alone and turns it all into work: a K–P violation.

Exam tip: Equivalence is proved by such combinations; both statements say the same thing in different words.

Q2 Answer: 1200

Concept: For a reversible engine $Q_2/Q_1 = T_2/T_1$, so $\eta = 1 - T_2/T_1$.

$\eta = 1 - 300/900 = 0.6667$

$W = 0.6667 \times 1800 =$ **1200 kJ** (heat rejected = 600 kJ)

Exam tip: $Q \propto T$ only for reversible devices; for any real engine the heat rejected is larger.

Q3 **Answer: 384.6 (accept 382.6 to 386.6)**

Concept: The minimum power corresponds to a reversible (Carnot) refrigerator.

$COP_{max} = T_L/(T_H - T_L) = 260.15/50 \approx 5.2$

$P_{min} = 2000/5.2 =$ **384.6 W**

Exam tip: Convert both temperatures to kelvin; COP uses the absolute temperature of the cold space.

Q4 **Answer: (C) 800 K**

Concept: For each reversible engine $\eta = 1 - T_{low}/T_{high}$.

Equal efficiencies: $T/1600 = 400/T$

$T = \sqrt{(1600 \times 400)} =$ **800 K** (each engine has $\eta = 50$ %)

For equal WORK instead, T would be the arithmetic mean, $(1600 + 400)/2 = 1000$ K.

Exam tip: Equal efficiency → geometric mean; equal work → arithmetic mean.

Q5 **Answer: 0.125 (accept 0.124 to 0.126)**

Concept: $S_{gen} = \Delta S_{block} + \Delta S_{lake}$; the lake is a reservoir, so $\Delta S_{lake} = Q/T_{lake}$.

$\Delta S_{block} = mc \ln(T_2/T_1) = 2 \times 0.4 \times \ln(300/500) = -0.4087$ kJ/K

Heat to lake $= mc(500 - 300) = 160$ kJ, so $\Delta S_{lake} = 160/300 = 0.5333$ kJ/K

$S_{gen} = -0.4087 + 0.5333 =$ **0.125 kJ/K**

Exam tip: The block's entropy falls; only the total (block + lake) must rise.

Q6 **Answer: (A) irreversible but possible**

Concept: $\oint \delta Q/T = 0$ for a reversible cycle, < 0 for an irreversible one, > 0 for an impossible one.

$\oint \delta Q/T = 500/1000 - 300/500 = 0.5 - 0.6 = -0.1$ kJ/K < 0

So the cycle is possible but irreversible. Check: $\eta = 200/500 = 40$ % $< \eta_{Carnot} = 50$ %.

Exam tip: Heat rejected enters the sum with a negative sign.

Q7 **Answer: 0.789 (accept 0.785 to 0.793)**

Concept: Ideal gas: $\Delta s = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$.

Rigid tank: $v_2 = v_1$, so the second term is zero.

$\Delta s = 0.718 \times \ln 3 =$ **0.789 kJ/kg·K**

Exam tip: Use c_v with the volume form and c_p with the pressure form of the entropy equation.

Q8 **Answer: (D) A is the constant-volume line, because its slope T/c_v is greater than T/c_p**

Concept: From $T\,ds = c_v\,dT + p\,dv$ and $T\,ds = c_p\,dT - v\,dp$:

$(\partial T/\partial s)_v = T/c_v$ and $(\partial T/\partial s)_p = T/c_p$.

Since $c_p > c_v$, the constant-volume line is steeper: **A is constant volume**, B is constant pressure.

Both curves are exponential, $T \propto e^{s/c}$.

Exam tip: On T–s, isochores are steeper than isobars; on p–v, adiabats are steeper than isotherms.

Q9 **Answer: 0.848 (accept 0.844 to 0.852)**

Concept: η_T = actual work / isentropic work = $(T_1 - T_2)/(T_1 - T_{2s})$ for an ideal gas with constant c_p.

$T_{2s} = 1000 \times (100/800)^{0.4/1.4} = 552.0$ K

$\eta_T = (1000 - 620)/(1000 - 552.0) = 380/448.0 =$ **0.848**

Exam tip: For a turbine the actual exit is hotter than isentropic; for a compressor the actual exit is also hotter, but η = isentropic/actual work.

Q10 **Answer: (B) 1000 kJ**

Concept: On a T–S diagram the area enclosed by a reversible cycle equals the net work (= net heat).

Heat in (1–2) = 800 × (3 − 1) = 1600 kJ; heat out (3–4) = 300 × 2 = 600 kJ

$W_{net} = (800 - 300) \times 2 =$ **1000 kJ**; η = 1000/1600 = 62.5 % = 1 − 300/800 ✓

Exam tip: A Carnot cycle is always a rectangle on T–s, whatever the working fluid.

Q11 **Answer: 350**

Concept: Chain the two reversible devices through the common work W.

W = (1 − 300/1000) × 100 = 70 kJ

$COP_R = 250/(300 - 250) = 5$

$Q_3 = 5 \times 70 =$ **350 kJ** (atmosphere receives 30 + 420 = 450 kJ)

Exam tip: The refrigerator can remove more heat than the source supplies, because the atmosphere absorbs it all.

Q12 **Answer: (C) Slow, frictionless adiabatic expansion of a gas behind a piston**

Concept: A process is reversible only if it is quasi-static, frictionless and involves no heat transfer across a finite temperature difference.

Throttling (A) involves friction; mixing of two gases (B) is spontaneous; stirring work (D) is dissipated by friction.

Slow, frictionless adiabatic expansion is the isentropic process: **reversible**.

Exam tip: Paddle-wheel (stirring) work can be done ON a system but never extracted, so it is always irreversible.

Q13 **Answer: 0.462 (accept 0.459 to 0.465)**

Concept: Throttling: $h_2 = h_1$. For an ideal gas h = h(T), so $T_2 = T_1$.

$\Delta s = c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = 0 - 0.287 \ln(0.2/1) = 0.4619$ kJ/kg·K

The valve is adiabatic, so $s_{gen} = \Delta s =$ **0.462 kJ/kg·K**

Exam tip: Throttling is isenthalpic, not isentropic; entropy always rises.

Q14 **Answer: (A) 0.606 W/K**

Concept: Steady state: the wall's entropy is constant, so S_{gen} = entropy out − entropy in.

$S_{gen} = 2000/275 - 2000/300 = 7.273 - 6.667 =$ **0.606 W/K**

Exam tip: Take the boundary temperatures of the system you choose; a wider boundary (room air to outside air) gives a larger S_{gen}.

Q15 **Answer: (D) 4**

Concept: $COP_{HP} = Q_H/W$ and $COP_R = Q_L/W$, with $Q_H = Q_L + W$.

$COP_R = COP_{HP} - 1 = 5 - 1 = \mathbf{4}$

Check: $T_H/(T_H - T_L) = 5$ gives $T_L/T_H = 0.8$, so $COP_R = 0.8/0.2 = 4$ ✓

Exam tip: The '+1' is just the work itself, which ends up in the hot reservoir.

What next?

Day 38 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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